SCIENCE · THE BIBLE · YOGA

A Tireless Machine Within, Labouring Hands Without

The Secret of the Body in Science, Scripture and Yoga



Neelam P.J. Pradeep Jesudas, B.Th

JAI BHIM EDUCATIONAL CHARITABLE TRUST

SCIENCE · THE BIBLE · YOGA

A Tireless Machine Within, Labouring Hands Without

The Secret of the Body in Science, Scripture and Yoga

✦ ✦ ✦

AUTHOR

Neelam P.J. Pradeep Jesudas

B.Th (Bachelor of Theology)

Writer, Activist and Social Worker

Published by

Jai Bhim Educational Charitable Trust

A Tireless Machine Within, Labouring Hands Without

The Secret of the Body in Science, Scripture and Yoga

Author: Neelam P.J. Pradeep Jesudas, B.Th
Writer, Activist and Social Worker

© Neelam P.J. Pradeep Jesudas. All rights reserved by the author.

Published by: Jai Bhim Educational Charitable Trust

English edition, translated from the Tamil original *Ulle Oyaatha Iyanthiram, Veliye Uzhaikkum Karangal*.

Scripture quotations follow the King James Version.

Illustrations: original digital artwork created for this book.

This book is not medical advice; anyone with a health condition should consult a doctor. It is a respectful comparative study, with no intention of belittling any tradition.

Preface

One body, two kinds of work, three ways of seeing

At four in the morning the fishermen of Thoothukudi walk toward the harbour. In the salt pans, women rake salt under a fierce sun. Young men pedal their bicycles to building sites. Their hands and feet labour for eight, ten, twelve hours. Yet another labour goes on inside them, one they never notice: the heart beats, the lungs breathe, the kidneys filter the blood, the liver transforms poisons. This inner labour has no holiday, no Sunday, no night.

The seed of this book is a simple thought that has run through my mind for many years: the human body is a single machine, yet two kinds of work go on within it. Inside, without asking our permission, organs labour without rest twenty-four hours a day. Outside, at our command, hands and feet labour for a few hours each day. When the hands outside work honestly, the machine inside receives what it needs. Outer labour is nourishment for the inner. This book tries to show that this is not merely a metaphor, and that modern exercise science confirms it.

We look at this idea through three lenses. The first is modern science: physiology, neuroscience, exercise science and sleep science. Here I have taken only well-established facts, and where numbers are approximate I have said so plainly. The second is the Bible: the deep theology of the body that runs from Genesis to Revelation. The third is the yogic tradition: karma yoga, prana, pranayama, asana, yoga nidra, and Tirumular's vision that "the body is a temple." I have tried to present these through their own texts, with respect and with accuracy.

I am not the first traveller on this road. Paul Brand, the surgeon who served leprosy patients at Vellore, together with the writer Philip Yancey, wrote *Fearfully and Wonderfully Made* (1980), which spoke beautifully of the body's cells, bones and skin as a parable of the Church, the body of Christ. The Benedictine monk J.-M. Déchanet, in *Christian Yoga*, wrote of a Christian monk's attempt to join yogic

practice to a life of prayer. The science journalist William J. Broad, in *The Science of Yoga* (2012), examined honestly which claims about yoga are confirmed by research and which are exaggerated.

What, then, does this book add? Three things. First, it seats all three traditions at one table around a single twofold structure: involuntary inner movement and voluntary outer movement. Second, it is written on Tamil soil, in the language of working people; here Tirumular and Valluvar converse with the Bible. Third, it does not stop at health advice. It faces the hard truth that the body breaks, falls ill and finally dies, and it ends with the answer the Christian faith gives to that breaking in Jesus Christ.

A word of caution: this is not a medical book. Anyone with an illness should begin exercise, a change of diet, fasting or breathing practice only on their doctor's advice. This book is an invitation to marvel at the body, to give thanks for it, and to seek the One who gave it. My prayer is that you will see the handiwork of the one Creator both in the sweat of the salt-pan worker and in the unceasing beat of the heart.

With love,

Neelam P.J. Pradeep Jesudas, B.Th

Author

Contents

Preface 4

CHAPTER 1 7
One Body, Two Movements

CHAPTER 2 13
The Tireless Inner Organs

CHAPTER 3 19
Labouring Hands

CHAPTER 4 25
Outer Labour Feeds the Inner

CHAPTER 5 31
The Secret of Rest

CHAPTER 6 37
Breath

CHAPTER 7 43
Food and Fasting

CHAPTER 8 49
One Body, Many Members

CHAPTER 9 55
The Body Is a Temple

CHAPTER 10 62
The Creator's Handiwork

CHAPTER 11 68
The Broken Machine and the Redeemer

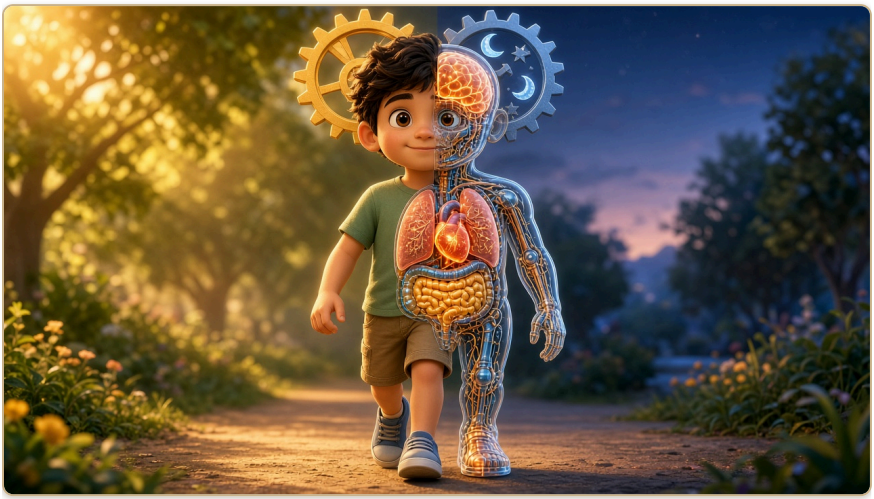
CHAPTER 12 74
The Risen Body

Sources 80

Acknowledgements 83

CHAPTER 1

One Body, Two Movements



Inside, a tireless machine; outside, labouring hands.

1.1 A Fisherman's Day

Antony is an imagined fisherman, but on the Thoothukudi shore there are a thousand like him. At dawn he pushes off his boat, casts his net, hauls in the heavy net, reaches the shore and lifts the baskets of fish. His hands, shoulders and legs move at his command. He thinks “pull,” and the muscles pull. He thinks “stop,” and they stop.

Yet at the same moment, work he never thinks of for an instant goes on inside him. As he hauls the net his heart beats faster; his lungs draw in more air;

more blood is redirected to his muscles; as his body heats up, sweat flows to cool it. He gave no order for any of this. Even if he falls asleep in the boat, none of it stops. Two worlds in one body: a world that obeys command, and a world that labours with no command at all.

1.2 The Two Divisions of the Nervous System

Physiology has given these two worlds names. The skeletal muscles that move at our will are governed by the *somatic nervous system*. A command from the brain travels down the spinal cord and, through a single nerve, reaches the muscle directly and makes it contract.

The inner organs that run without our will are governed by the *autonomic nervous system*. It has two branches. One is the *sympathetic* system: in danger and in labour it raises the heart rate, widens the airways and prepares the body to "fight or flee." The other is the *parasympathetic* system: at rest and after meals it slows the heart, stimulates digestion and restores the body to "rest and digest." Its chief nerve, the *vagus nerve*, runs from the brainstem down to the heart, lungs, stomach and intestines.

The heart muscle, and the smooth muscle of the gut and blood vessels, do not move at our will. The heart has its own natural pacemaker, the *sinoatrial node*; the nerves change its speed, but it is the pacemaker that starts each beat. That is why a transplanted heart beats even though its nerves have been cut.

1.3 Where the Two Meet: the Breath

Between these two worlds there is a beautiful bridge: breathing. When we sleep, breath goes on by itself; when we are awake we can slow it, stop it, or deepen it at will. The diaphragm is a skeletal muscle, yet the centres in the brainstem that drive it keep it moving even when we forget.

The yogic tradition discovered this bridge centuries ago. Pranayama is the art of handling, with awareness, an action that otherwise happens on its own.

Today's research shows that slow breathing stimulates the parasympathetic system and calms the heart; we shall look at this in detail in Chapter 6. It is at the place where will and the involuntary meet that a human being begins to converse with his own body.

✦ Did you know?

If a heart is removed from the body and kept in a suitable fluid, it will go on beating for a while with no nerve connection at all. Because the nerves of a transplanted heart have been cut, at rest it usually beats somewhat faster than normal: it no longer has the vagus nerve's "brake."

1.4 The Bible: Dust and Breath

The Bible presents the very beginning of humanity as a twofold movement:

> "And the LORD God formed man of the dust of the ground, and breathed into his nostrils the breath of life; and man became a living soul." (Genesis 2:7)

On one side, a body made of dust; on the other, the breath of God. The body received its form, but it received life only when God breathed. The believer sees even today our inner movements, the beating and breathing that happen without our will, as the continuation of that first breath. "For in him we live, and move, and have our being" (Acts 17:28).

In the same chapter God also gives man outer work: "And the LORD God took the man, and put him into the garden of Eden to dress it and to keep it" (Genesis 2:15). Inside, the life God gave; outside, the work God gave. Before sin entered, from the very beginning, these two were together. Labour is not a curse; it is part of creation.

Paul speaks of the same pair in another way: "though our outward man perish, yet the inward man is renewed day by day" (2 Corinthians 4:16). He is speaking not of physiology but of the spiritual life; yet it is a deep echo of this book's "inside and outside" structure.

1.5 Yoga: the Five Sheaths

The Taittiriya Upanishad describes the human being as five *koshas*, or sheaths: the *annamaya kosha* made of food (the body), the *pranamaya kosha* made of prana, the *manomaya kosha* made of mind, the *vijnanamaya kosha* made of understanding, and the *anandamaya kosha* of bliss. Yoga teachers make wide use of this scheme.

Closest to the theme of this book is the pranamaya kosha. Tradition says that the prana which gives life to the body divides into five *vayus*, or winds, that carry out the inner movements: *prana*, which drives breathing; *apana*, which expels waste; *samana*, which drives digestion; *udana*, which drives speech and upward movement; and *vyana*, which circulates through the whole body. This is not the language of modern physiology, yet it expresses the same wonder: within us an unceasing life-force does many kinds of work.

All three views circle around one truth: a human being is greater than the body he controls; within him runs a life he does not control. Science calls it the nervous system; yoga calls it prana; the Bible calls it the breath of life that God breathed in.

1.6 Not a Machine Without a Driver

Some modern thinkers say the human body is "only a biological machine." In the seventeenth century the philosopher René Descartes saw the bodies of animals as machines, and said that only man has a thinking mind separate from the body. Some who came after him made that mind, too, part of the machine.

Because the word “machine” appears in the title of this book, this must be made clear: here “machine” is only a metaphor.

Someone must make a clock, and someone must wind it. But the body grows itself, repairs itself and adapts to its surroundings; no machine does this. Moreover, within that “machine” a person loves, grieves, prays and gives himself for others. The Bible does not split man into a separate body and a separate soul; it sees him as an integrated life, a “living soul” (*nephesh*). The yogic tradition too sees the body as the vehicle of the soul, a “chariot” (Katha Upanishad): there is a traveller in the chariot.

So when this book calls the body a machine, it marvels at its order and reliability; it does not lessen its dignity. Where the metaphor of the machine ends, the mystery of the human being begins.

1.7 The Journey of This Book

This first chapter is a map. The next four chapters explore the body's two movements and the relationship between them: the tireless inner organs (2), the labouring hands (3), the nourishment outer labour gives the inner (4), and the secret of rest (5). Then three chapters go into the daily life of the body: breath (6), food and fasting (7), and the unity of many members (8). The last four chapters ask deeper questions: is the body holy (9)? Who designed it (10)? Why does it break, and is there redemption for it (11)? What happens to the body after death (12)?

In every chapter three voices speak. The voice of science asks, “How?” The voice of the yogic tradition asks, “How do we practise?” The voice of the Bible asks, “For what? For whom?” To give room to all three voices is the honesty of this book; to listen in the end to the voice of the Bible is the faith of its author.

For Reflection

1. Can you list the work that went on in your body today without any command from you? Have you ever given thanks for it?
2. "Labour is not a curse; it is part of creation" (Gen 2:15). How does this change the way you see your own work?
3. Breath is the bridge between will and the involuntary. Breathe slowly for one minute; what changes inside?

Comparison

	Science	The Bible	Yogic tradition
Inner movement	Autonomic nervous system (sympathetic, parasympathetic)	The breath of life God breathed (Gen 2:7)	The five vayus of prana
Outer movement	Somatic nervous system, skeletal muscles	The task of tending the garden (Gen 2:15)	Karma, asana
The bridge	Breathing: involuntary and voluntary	Breath and Spirit (ruach)	Pranayama
Who is man?	An integrated biological system	A living soul of dust and breath	The five koshas

✦ ✦ ✦

CHAPTER 2

The Tireless Inner Organs



Sleepless guardians: heart, lungs, kidneys, liver, gut.

2.1 The Guardian Who Never Sleeps

Have you ever thought, as you sleep at night, that someone ought to stay awake to guard your house? Inside your body such guardians never sleep. The psalmist sings of God:

> "Behold, he that keepeth Israel shall neither slumber nor sleep." (Psalm 121:4)

This verse is not about our inner organs; it is about God's keeping hand. Yet when we think of the heart that beats unsleeping inside a sleeping man, we may say that we glimpse a small likeness of that keeping hand in our own body. In this chapter we shall meet five "sleepless guardians."

2.2 The Heart: a Muscle That Beats for a Lifetime

The heart of a resting adult beats about 60 to 100 times a minute. At 70 a minute, that is about a hundred thousand beats a day, and more than two and a half billion in seventy years. Every minute at rest it pushes about five litres of blood through the whole body; in hard exercise this rises several-fold.

Does the heart never rest? Between each beat there is a short pause, *diastole*; then the chambers relax and fill with blood, and most of the flow in the coronary vessels that feed the heart itself takes place. Rest is woven into the work itself. This is a prelude to the lesson of rest in Chapter 5.

The Bible seldom uses the word "heart" for the physical organ; it means a person's inner centre, thought and will. Yet this verse rings true in both senses: "Keep thy heart with all diligence; for out of it are the issues of life" (Proverbs 4:23).

2.3 Lungs, Kidneys, Liver

A resting adult breathes about 12 to 20 times a minute, roughly twenty thousand breaths a day. Inside the lungs are hundreds of millions of air sacs, the *alveoli*; textbooks describe their total surface as a good part of a tennis court (estimates vary from about 50 to 100 square metres). Through that thin wall oxygen passes into the blood and carbon dioxide passes out.

Each of the two kidneys is only the size of a fist. Yet together they filter about 180 litres of fluid a day, reabsorb 99 per cent of it, and pass only about one to two litres as urine. They watch over salt, water, acid-base balance and blood pressure.

The liver is the largest internal organ of the body; it performs hundreds of chemical tasks. It processes nutrients from food, stores glucose as glycogen and releases it when needed, transforms drugs and toxins, secretes bile, and makes the proteins the blood needs to clot. It has the rare power to grow back even when part of it is removed.

2.4 The Gut: a Second Brain

In the walls of the digestive tract lies a separate web of nerves, the *enteric nervous system*. It contains hundreds of millions of neurons; estimates range from about two hundred to six hundred million. It can drive the wave-like movement of the gut, *peristalsis*, with no command from the brain, which is why some call it the "second brain." Yet it is also in constant conversation with the brain through the vagus nerve.

The countless microbes living in the gut, the *gut microbiota*, help digestion and immunity. Present estimates say their number is roughly equal to the number of the body's own cells (the old figure of "ten times as many" was corrected as an exaggeration in 2016). The inner lining of the gut wall is renewed every few days. All of this goes on without asking our permission.

✦ Did you know?

Walter Cannon, who coined the word "homeostasis," also popularised the phrase "fight or flight." The same scholar who described the body's inner wisdom also described how suddenly that wisdom shifts in fear.

2.5 Yoga: the Inner Fire and the Inner Wind

The yogic and Ayurvedic traditions call digestion *jatharagni*, the fire of the belly, and say that samana vayu tends that fire. Hatha yoga texts claim that abdominal practices such as *nauli* and *uddiyana bandha* stimulate the inner

organs. Some of these claims have not been tested by modern research, and that must be said honestly.

But the basic intuition of this tradition deserves respect: the inner organs are not separate machines; they are many expressions of one life-force, each dependent on the others. Modern science is moving in the same direction: the gut-brain axis, the heart-kidney link, the metabolic conversation between liver and muscle. The body is an orchestra; each instrument plays its own part, but the song is one.

2.6 We Know Only When We Fall Ill

We mostly learn the worth of our inner organs only when they falter. Go and look at the dialysis unit of a government hospital in Thoothukudi. Two or three times a week, up to four hours each time, people are connected to a machine and their blood is cleansed. The work that two small kidneys did quietly night and day without our permission, a large machine can do only part-time, and even then incompletely.

For a heart patient every flight of stairs is a mountain. For one who is breathless every breath is a struggle. Only then do we realise that health is a quiet miracle; it makes no noise. On the days when we are well, we never notice that miracle at all.

David sang: "Bless the LORD, O my soul, and forget not all his benefits: Who forgiveth all thine iniquities; who healeth all thy diseases" (Psalm 103:2–3). The word "forget not" matters. Gratitude is an exercise of memory. If, on waking each morning, we learn to say, "Tonight too my heart beat and my kidneys worked; thank you," we learn gratitude before illness comes.

2.7 The Wisdom of Balance

In the 1930s the American physiologist Walter Cannon, in *The Wisdom of the Body* (1932), gave the name *homeostasis* to the body's power to keep its inner conditions (temperature, sugar, salt, acidity) steady within narrow limits. Earlier still, the French physiologist Claude Bernard had put forward the idea of the "inner environment," the *milieu intérieur*. Whether the sun blazes outside or the rain pours, the body's inner temperature stays close to about 37° Celsius.

When the believer looks at this wisdom he remembers the words of Job: "The Spirit of God hath made me, and the breath of the Almighty hath given me life" (Job 33:4). The unceasing labour of the inner organs is not our achievement; it is a gift received each day. Every time we open our eyes in the morning, the beating heart reminds us of the verse, "They are new every morning" (Lamentations 3:23).

2.8 The Cost of Unceasing Labour

The inner organs need energy to labour without rest. The energy the body spends even at rest is called the *basal metabolic rate*; for most people it is the greater part of a day's total energy use. The brain is only about two per cent of body weight; yet textbooks say that at rest it uses roughly a fifth of the body's energy. The liver, kidneys and heart take another large share.

In other words, even when we lie doing nothing our body labours hard. The "wage" we pay for this labour is food, oxygen and water. Here the central question of this book begins: how do we, on the outside, give those who labour unceasingly on the inside what they need? The next two chapters seek the answer. Jesus' word applies to the body too: "the labourer is worthy of his hire" (Luke 10:7).

For Reflection

1. What comfort does the verse "shall neither slumber nor sleep" give to someone who lies down at night full of worry?
2. Which of your inner organs did you learn something new about today? What in your habits should it change?
3. How does the image of the body as an orchestra apply to a family or a community?

Comparison

	Science	The Bible	Yogic tradition
Heart	About 60–100 beats a minute at rest	"Keep thy heart" (Prov 4:23)	Vyana vayu
Lungs	About 12–20 breaths a minute at rest	The breath of life (Gen 2:7)	Prana vayu
Kidneys	About 180 litres filtered a day	—	Apana vayu
Gut	Enteric nervous system, microbes	—	Samana vayu, jatharagni
Integration	Homeostasis (Cannon)	"The keeper who never sleeps" (Ps 121:4)	One prana

✦ ✦ ✦

CHAPTER 3

Labouring Hands



The carpenter's hands, the farmer's hands, a mother's hands.

3.1 The Wonder of the Hand

Look closely at your hand for a moment. Each hand has 27 bones (the count may vary slightly by method); together the two hands hold nearly a quarter of the body's 206 bones. Many of the muscles that move the fingers actually lie in the forearm, pulling the fingers through long tendons, as a puppeteer pulls strings. The opposable thumb lets us thread a needle and grip a heavy hammer.

The famous “homunculus” drawn by the neurosurgeon Wilder Penfield shows that the space in the brain's *motor cortex* given to the hands, fingers and

face is out of all proportion to their size. The human brain is, in a sense, largely devoted to the working hand.

Dr Paul Brand spent most of his life restoring the hands of leprosy patients. The tendon-transfer operations he developed at Vellore gave working hands back to many whose fingers had curled up. He also discovered that hands are injured because leprosy destroys the sense of pain. To be able to work with one's hands again is to receive one's dignity again.

3.2 "In the Sweat of Thy Face"

The words God spoke after Adam's sin are often misunderstood:

> "In the sweat of thy face shalt thou eat bread, till thou return unto the ground; for out of it wast thou taken: for dust thou art, and unto dust shalt thou return." (Genesis 3:19)

The curse here is not on labour. As we saw in the first chapter, work was given before sin (Gen 2:15). The curse is on the ground: "cursed is the ground for thy sake ... Thorns also and thistles shall it bring forth to thee" (3:17–18). Joyful work now became sweat and struggle. Labour is still good; but it takes place in a broken world.

Today's physiology adds a new meaning to this "sweat." Sweat is the body's cooling system; it carries away the heat produced by working muscles. Even the sweat of a man labouring on cursed ground is a system of grace that protects him.

3.3 "If Any Would Not Work"

In Thessalonica some Christians said, "Christ is coming soon; there is no need to work any more," and they gave up work and lived off others. Paul wrote:

> "For even when we were with you, this we commanded you, that if any would not work, neither should he eat." (2 Thessalonians 3:10)

Notice: not "the one who cannot work" but "the one who will not." This verse is not about the sick, the elderly, the differently abled or those who cannot find work. The same Paul insists elsewhere that caring for them is the Church's duty. It is about the one who is able to work and chooses idleness.

Paul held himself up as an example. He was a tentmaker by trade (Acts 18:3). Showing his hands to the elders of Ephesus he said, "these hands have ministered unto my necessities, and to them that were with me" (Acts 20:34). Jesus himself was called by the people of his town, "Is not this the carpenter?" (Mark 6:3). The Word through whom the world was made lived in Nazareth with hands that shaped wood and stone.

✦ Did you know?

The Greek word Mark 6:3 uses for Jesus is *tektōn*. Many scholars think it can mean not only a woodworker but a builder who works in both stone and wood. The city of Sepphoris near Nazareth was being rebuilt at that time; that the young Jesus may have worked there is only a conjecture.

3.4 Karma Yoga: Action Without Attachment to Its Fruit

The third chapter of the Bhagavad Gita is called "Karma Yoga." When Arjuna wishes to abandon action on the battlefield, Krishna says:

> "Niyatam kuru karma tvam karma jyayo hy akarmanah | sharira-yatrapi cha te na prasiddhyed akarmanah ||" — Do the action assigned to you; action is better than inaction; without action even the journey of your body cannot go on. (Gita 3:8)

"Even the journey of your body cannot go on" (*śarīra-yātrā*): how close this line is to the central idea of this book! Another famous line of the Gita: "Karmany evadhikaras te ma phaleshu kadachana" — you have a right to the action alone, never to its fruits (2:47). And again: "Yogah karmasu kaushalam" — yoga is skill in action (2:50).

In his lectures published as *Karma Yoga* in 1896, Swami Vivekananda explained that work done without expecting its fruit purifies the mind. Labour is a spiritual discipline; liberation comes through action itself.

3.5 Karma Yoga and Christian Labour

One verse of Paul resonates with karma yoga in a striking way:

> "And whatsoever ye do, do it heartily, as to the Lord, and not unto men." (Colossians 3:23)

Both turn labour into a spiritual state. Neither makes human praise or reward the centre. But there is a subtle difference. In the Gita, action done without attachment to its fruit becomes the way to freedom from the bondage of karma. For Paul, labour is not the way to earn salvation; salvation has already been given by grace (Ephesians 2:8–9). Labour is the grateful response to that grace: "For we are his workmanship, created in Christ Jesus unto good works" (Ephesians 2:10).

Valluvar too bows before the labouring hand: "Uzhuvaar ulagaththaarkku aani; aqdhaatraadhu ezhuvaarai ellaam poruththu" (Kural 1032): the ploughmen are the linchpin of the world; they bear up all those who cannot plough and turn to other trades. The hands that shed sweat in the salt pans, on the sea and in the fields are the linchpin of society. A spirituality that does not honour those hands is not true spirituality.

3.6 The Weariness and the Dignity of Hands

Not all labour is joyful. The labour of people who go down into mines, match factories and sewers is often a burden of injustice. The degrading practice of manual scavenging has not been wholly abolished even after being banned by law; most of those who die going down into septic tanks belong to oppressed communities. To speak to them of "labour as a spiritual discipline" is cruel mockery.

The Bible is not silent about this injustice. James writes: "Behold, the hire of the labourers who have reaped down your fields, which is of you kept back by fraud, crieth: and the cries of them which have reaped are entered into the ears of the Lord of sabaoth" (James 5:4). Deuteronomy commands that wages be paid the same day (24:14–15). Whoever speaks of the dignity of labour must also speak of the rights of the labourer.

It must also be said honestly that the Gita's idea of *svadharma* (one's own duty) has at times been used to justify a caste system that fixes occupation by birth; many Hindu thinkers have reinterpreted it in terms of temperament and ability. Dr Ambedkar criticised caste as "not a division of labour but a division of labourers." If labouring hands have dignity, no hand must be bound by birth to degrading work. This is the social conclusion of this chapter.

3.7 The Prayer of Folded Hands

Labouring hands also become praying hands. The drawing *Praying Hands* made by the German artist Albrecht Dürer in 1508 is world-famous: two hands, worn by labour, veins standing out, joined and raised upward. A popular story says the hands were those of Dürer's brother, who laboured in the mines to pay for Dürer's studies; but historians regard it as a later invention. Even if the story is invented, the truth it touches is real: labour and prayer meet in the same hands.

The old motto of the Benedictine monks is *Ora et labora*: pray and work. In the Hindu tradition too there is a custom of joining the hands in worship before beginning work; the morning śloka recited while looking at the palms, "Karatale vasate Govindah," sees God in the very hands. In both traditions hands do two kinds of work: they shape the world, and they rise toward God.

For Reflection

1. If you did your work "as to the Lord," what would change at your workplace tomorrow?
2. Can you put into your own words the difference between karma yoga's "non-attachment to the fruit" and the Christian view of work as a "grateful response" to grace?
3. Whose labouring hands are the "linchpin" of society in your town? How do we honour them?

Comparison

	Science	The Bible	Yogic tradition
The hand	27 bones, a large area in the brain	"These hands" (Acts 20:34)	Karmendriya (organ of action)
Origin of labour	Evolution, tool use (scientific account)	A task given before sin (Gen 2:15)	Assigned karma (Gita 3:8)
Sweat	Body temperature regulation	"In the sweat of thy face" (Gen 3:19)	Tapas (the heat of effort)
Purpose of labour	Physical and mental health, livelihood	As to the Lord (Col 3:23)	Unattached action (Gita 2:47)
Idleness	Physical harm of inactivity	"Neither should he eat" (2 Thess 3:10)	Tamas

CHAPTER 4

Outer Labour Feeds the Inner

✦



When the hands labour, heart, blood, sleep and mind grow strong.

4.1 The Lesson of the London Buses

In 1953 a British epidemiologist named Jeremy Morris published in *The Lancet* a simple but historic study. London's double-decker buses had two kinds of staff: drivers who sat all day, and conductors who climbed up and down the stairs all day selling tickets. Same company, same city, much the same

social background. Yet the conductors had less coronary heart disease, and when it came, it came less severely.

In the same study he compared postal clerks who worked seated in offices with postmen who delivered letters on foot, and the same pattern appeared. Many regard this study as the moment the scientific world began to notice the link between physical labour and heart health. Later, other factors such as body size and stress were examined too; but the basic message has been confirmed by thousands of studies: a body that moves protects its heart.

4.2 The Heart and the Blood Vessels

With regular exercise the heart muscle grows stronger and can push more blood with each beat. So its resting rate usually falls. Well-trained long-distance runners may have resting heart rates even in the forties. Outer labour earns rest for the inner machine!

The inner lining of the blood vessels, the *endothelium*, is stimulated by the increased blood flow of exercise to produce a substance called nitric oxide, which widens the vessels. Regular exercise helps many people lower their blood pressure. When the leg muscles contract they squeeze the veins and push blood upward toward the heart; this is called the "muscle pump." One reason people who stand still for long hours get swollen legs is that this pump is not working. Walking legs are, in a sense, a second heart.

4.3 Muscle: the Organ That Eats Sugar

In India diabetes is growing into a great burden. Here we see the most important inner benefit of outer labour. After a meal the body's skeletal muscles take up most of the glucose in the blood. Normally this needs insulin: insulin brings "glucose doors" called GLUT4 to the surface of muscle cells.

But it is a well-established finding that when muscle contracts, the same doors open even without insulin. That is why a short walk after a meal helps

blunt the peak of blood sugar. Regular exercise increases *insulin sensitivity*, meaning the same amount of insulin does more work. Exercise is a basic part of medical guidelines for preventing and controlling type 2 diabetes. (People with diabetes should coordinate medicine and exercise on their doctor's advice.)

In the past twenty years researchers such as Bente Klarlund Pedersen of Denmark have shown that contracting muscle secretes signalling proteins called *myokines*. Muscle, in other words, also acts like an endocrine gland; it speaks in chemical language with inner organs such as the liver, fat tissue and brain. When the hand labours, it writes letters to the inner organs!

4.4 Sleep and the Mind

"The sleep of a labouring man is sweet, whether he eat little or much: but the abundance of the rich will not suffer him to sleep" (Ecclesiastes 5:12). Today's research largely supports this practical wisdom of three thousand years ago: many studies show that regular exercise improves the quality of sleep (though hard exercise just before bed can have the opposite effect for some).

The same is true for the mind. Reviews of many studies show that exercise helps reduce the symptoms of depression. Studies suggest it raises a growth factor in the brain called BDNF and supports the health of the hippocampus, a region vital for memory. "A merry heart doeth good like a medicine: but a broken spirit drieth the bones" (Proverbs 17:22). Body and mind are not two houses; they are two rooms of one house.

✦ Did you know?

Because the calf muscles push the blood in the veins upward as we stand and walk, doctors sometimes call them the "second heart." On long flights we are told to move our legs from time to time precisely to work this pump and lower the risk of blood clots.

4.5 How Much Is Enough?

According to the guidelines published by the World Health Organization (WHO) in 2020, adults should do 150 to 300 minutes of moderate physical activity a week (such as brisk walking), or 75 to 150 minutes of vigorous activity, along with muscle-strengthening exercise on two days a week. It also advises reducing time spent sitting. Its message, too, is that "doing some is better than doing none."

Here a question of social justice arises. The salt-pan worker and the construction worker have no lack of physical labour; what they need is enough rest, nourishing food, a safe workplace and a fair wage. Excessive labour without rest breaks the body. On the other hand, the young man who sits ten hours before a computer has no outer labour at all. One has too much, the other too little. This book's message differs for the two: to one, "rest is your right"; to the other, "rise and walk."

4.6 A Doctor's Prescription

Doctors often say that if exercise could be packed into a pill, it would be the best-selling medicine in the world. It strengthens the heart and blood vessels, controls sugar, helps preserve bone density, deepens sleep, improves mood and reduces the risk of falls in the elderly. Even bones respond to labour: weight-bearing exercise gives signals that strengthen bone, while long bed rest and the weightlessness of astronauts cause bone loss.

But this "medicine" also has a dose. Hard exercise begun suddenly can cause injuries, and danger for those with heart disease. Wisdom is to begin slowly and increase step by step. Thirty minutes of brisk walking a day, five days a week, reaches the minimum the WHO recommends.

Think of the traditional bodily arts of Tamil soil: *silambam*, *kabaddi*, *uriyadi*, drawing water with the *etram*, *kolattam*, *kummi*. There was a time when labour, play and song were not separated. Our grandfathers did not set

aside time for “exercise”; their life itself was exercise. In today's city life we must deliberately recover it.

4.7 “Bodily Exercise Profiteth Little”?

Some Christians quote this verse and dismiss exercise: “For bodily exercise profiteth little: but godliness is profitable unto all things, having promise of the life that now is, and of that which is to come” (1 Timothy 4:8). But Paul does not say “profits nothing”; he says “profits a little” (Greek: *pros oligon*). He does not reject exercise; he puts it in its proper place. The body's health is for this life; godliness is for both lives.

The yogic tradition speaks of the same balance. In Patanjali's Yoga Sutras asana is only one of eight limbs; the goal is not the body but the stilling of the mind: “Yogash chitta-vritti-nirodhah” (1:2). Both traditions honour the body, yet neither makes it a god. Exercise is a good servant and a bad master.

4.8 Inactivity, the New Disease

Doctors today warn a great deal about the *sedentary lifestyle*. The World Health Organization notes that insufficient physical activity is one of the leading risk factors for preventable death in the world. Some say, exaggerating, that “sitting is the new smoking”; the comparison is not accurate, but its warning deserves attention.

Some small studies show that for those who sit all day, getting up every half hour to walk for a few minutes may help reduce rises in blood sugar. It is a simple habit: using the stairs at the office, walking while talking on the phone, going to the nearby shop without a vehicle.

The writer of Proverbs sends the sluggard to the ant: “Go to the ant, thou sluggard; consider her ways, and be wise” (Proverbs 6:6). The lesson the ant gives today's young man: get up, walk, work; your inner machine will thank you.

For Reflection

1. The London bus driver or the conductor: which does your daily life resemble?
2. Who in your community is "broken by too much labour" and who is "weakened by too little"? What can the Church and society do for each?
3. How does 1 Timothy 4:8 balance bodily exercise and godliness?

Comparison

	Science	The Bible	Yogic tradition
Heart	Lower resting rate, lower risk of heart disease	"Keep thy heart"	Asana, Surya Namaskar
Sugar	GLUT4 through muscle contraction, insulin sensitivity	—	Moderate food and effort (Gita 6:17)
Sleep	Better quality	"The sleep of a labouring man is sweet" (Eccl 5:12)	Yukta cheshta (fitting action)
Mind	Fewer symptoms of depression, BDNF	"A merry heart doeth good like a medicine" (Prov 17:22)	Chitta-vritti-nirodha
Limit	WHO: 150–300 minutes a week	"Profiteth little" (1 Tim 4:8)	One of eight limbs

✦ ✦ ✦

CHAPTER 5

The Secret of Rest



The stillness of the seventh day and the sleep of the night.

5.1 The God Who Rested

On the first pages of the Bible stands a startling sentence: “And on the seventh day God ended his work which he had made; and he rested on the seventh day from all his work which he had made” (Genesis 2:2). It does not mean the Almighty grew tired. Of him we read that he “fainteth not, neither is weary” (Isaiah 40:28). He rested in order to give rest to humanity as a pattern and a gift.

Later it became a commandment: "Six days shalt thou labour, and do all thy work: but the seventh day is the sabbath of the LORD thy God" (Exodus 20:9–10). Notice who receives rest in that command: you, your son, your daughter, your manservant, your maidservant, your cattle, and the stranger within your gates. In the ancient world this was a revolutionary law that gave the right of rest to slaves and animals. Deuteronomy 5:15 gives the reason: "remember that thou wast a servant in the land of Egypt." Rest is the sign of freedom.

5.2 "The Sabbath Was Made for Man"

In Jesus' time the sabbath rules had become a heavy burden. To those who accused his disciples of plucking ears of grain in their hunger he said: "The sabbath was made for man, and not man for the sabbath" (Mark 2:27). It was on the sabbath that he healed many: the man with the withered hand, the man who had lain ill for thirty-eight years. Rest is a day for restoring the body; the witness he gave to it was healing.

Jesus' invitation is to body and soul together: "Come unto me, all ye that labour and are heavy laden, and I will give you rest" (Matthew 11:28). He himself sat down wearied by a well (John 4:6), slept in the boat in the middle of a storm (Mark 4:38), and said to his disciples, "Come ye yourselves apart into a desert place, and rest a while" (Mark 6:31).

5.3 The Science of Sleep

Nearly a third of our lives is spent asleep. It is not wasted time; it is maintenance time for the inner machine. Bodies such as the American Academy of Sleep Medicine recommend that adults sleep at least seven hours a night.

There are two main kinds of sleep: deep sleep without rapid eye movement (non-REM), and REM sleep, in which most dreams occur. Through the night they alternate in cycles of about 90 minutes. Studies show that during deep sleep more growth hormone is released, tissues are repaired, and what was

learned during the day is consolidated in memory. In 2013 a study in mice by Maiken Nedergaard's team showed that the "glymphatic" system, which clears the brain's waste, works more during sleep; its details in humans are still being studied.

Lack of sleep affects blood pressure, sugar control, immunity and mood. When the outer machine labours honestly, sleep deepens; deep sleep restores the inner machine; a restored inner machine gives strength for the next day's outer labour. It is a beautiful cycle.

✦ Did you know?

In the Hebrew calendar a day begins in the evening: "And the evening and the morning were the first day" (Gen 1:5). So in the biblical reckoning a person begins the day with sleep. Some Jewish and Christian writers have meditated that God works through the night, and man rises in the morning and joins in His work.

5.4 What Does the Inner Machine Do at Rest?

The parasympathetic nervous system we met in the first chapter is called "rest and digest." It comes to the fore when we sit quietly, after meals and in sleep. The heart rate falls; saliva and digestive juices flow more; the gut moves more. The level of cortisol, the stress hormone, usually falls at night and begins to rise in the early morning, preparing us to wake.

There is a biological clock, a *circadian clock*, within our body. A small cluster of neurons in the brain's hypothalamus, the *suprachiasmatic nucleus*, sets itself to the time of day by the light that falls on the eyes. In 2017 the Nobel Prize in Medicine was awarded to three American scientists for discovering the molecular workings of this inner clock. Studies suggest that the light of phone and computer screens at night can confuse this clock and delay sleep.

"And God made two great lights; the greater light to rule the day, and the lesser light to rule the night" (Genesis 1:16). Day and night are the rhythm of

creation. A body that walks with that rhythm flourishes; a body that walks against it grows weary. For night-shift workers this is a real challenge; it is society's duty to give them enough rest.

5.5 Yoga Nidra: Wakeful Rest

In the yogic tradition, *yoga nidra* is a state of deep relaxation with awareness. The form widely taught today was designed in the 1960s and 70s by Swami Satyananda Saraswati of the Bihar School of Yoga on the basis of older tantric practices. Its structure is to lie in shavasana, take a *sankalpa* (a resolve), move attention through each part of the body in turn, observe the breath, and pass through mental images.

Some small studies suggest that guided relaxation practices of this kind may help reduce stress and sleep problems; but large, high-quality studies are still few. Popular claims such as "one hour of yoga nidra equals four hours of sleep" have not been scientifically proven.

The Christian tradition also has wakeful rest. "Be still, and know that I am God" (Psalm 46:10). Yoga nidra attends to the body part by part; the Christian believer can at the same time give thanks for each member. The relaxation is the same; its centre is different.

5.6 Sleep, the Gift to the Beloved

The Bible presents sleep as a confession of faith: "I will both lay me down in peace, and sleep: for thou, LORD, only makest me dwell in safety" (Psalm 4:8). Another psalm looks at those who rise anxious at dawn and toil till midnight and says: "It is vain for you to rise up early, to sit up late, to eat the bread of sorrows: for so he giveth his beloved sleep" (Psalm 127:2).

Sleep is a daily surrender. When we sleep we control nothing; the outer machine stops entirely; we entrust the inner machine to One greater than ourselves. Every night we rehearse death in a small way; every morning,

resurrection in a small way. In the last chapter of this book this image will find its full meaning.

5.7 The Sleep of Peace

Sleeplessness has not only bodily causes; worry, guilt, fear and enmity also steal the night. When the sympathetic nervous system stays alert under stress, it is hard for the body to go into deep sleep. So good sleep also becomes a spiritual question.

Paul writes: “Be ye angry, and sin not: let not the sun go down upon your wrath” (Ephesians 4:26). Do not go to bed with enmity. Jesus' word, too, is medicine for the night: “Take therefore no thought for the morrow: for the morrow shall take thought for the things of itself” (Matthew 6:34).

The yogic *shavasana*, “the corpse pose,” is a deep symbol: the letting go of all effort. Its counterpart in the Christian tradition is night prayer; for centuries monks before sleeping said Jesus' word from the cross: “Father, into thy hands I commend my spirit” (Luke 23:46). The word first appears in Psalm 31:5. Every night is a small surrender.

For Reflection

1. Is there a real “sabbath” in your week? If not, what prevents it?
2. At home and at work, do you honour the right of workers to rest (Ex 20:10)?
3. Tonight before you sleep, say Psalm 4:8 and entrust your inner machine to God.

Comparison

	Science	The Bible	Yogic tradition
Rhythm of rest	Sleep cycles (about 90 minutes)	Seventh-day rest (Ex 20:9–10)	Shavasana, yoga nidra
Fruit of rest	Tissue repair, memory, waste clearance	"The sabbath was made for man" (Mark 2:27)	Calm of mind, sankalpa
Amount	Adults: at least 7 hours	—	Yukta svapna (fitting sleep, Gita 6:17)
Spiritual meaning	—	Sign of freedom, trust (Ps 4:8)	Wakeful relaxation
Caution	Harms of sleep loss	"Vain" toil (Ps 127:2)	Exaggerated claims

CHAPTER 6

Breath



From the first breath God breathed to every breath we take.

6.1 The Word for Breath

In Hebrew the word *ruach* means breath, wind and spirit. The Greek *pneuma* is the same. In Latin *spiritus*, from which English draws "spirit," "respiration" and "inspiration." In Sanskrit *prana* is the breath of life; some linguists think the word *atman* too is related to breath. In Tamil the word *uyir*, life, is joined to the verb *uyirttal*, to breathe. In many of the world's languages breath, life and spirit live in a single word. Humanity has always sensed that breath is something greater than mere air.

6.2 How the Lungs Move

The chief engine of breathing is the *diaphragm*, the dome-shaped muscle between chest and belly. When it contracts and moves down, the chest cavity enlarges and air comes in. When it relaxes, air goes out. The muscles between the ribs help too.

The surprise is that the main trigger for breathing is not a lack of oxygen but a rise of carbon dioxide in the blood. Sensors in the brainstem monitor it and adjust the rate and depth of breathing automatically. The compelling urge to breathe that rises when we hold our breath comes mostly from this rise in CO_2.

At rest about half a litre of air enters with each breath (the *tidal volume*). During exercise this increases many times over. Regular exercise strengthens the breathing muscles and increases the body's capacity to use oxygen (VO_2max). Studies show this capacity is one of the strongest measures linked to long life.

6.3 Pranayama: the Art of Breath

Patanjali's Yoga Sutras define pranayama thus: "Tasmin sati shvasa-prashvasayor gati-vicchedah pranayamah" — once posture is established, the regulation of the natural flow of in-breath and out-breath is pranayama (2:49). Later texts such as the Hatha Yoga Pradipika describe many methods: *nadi shodhana* (alternate-nostril breathing), *ujjayi*, *bhastrika*, *bhramari*. The idea that "when the breath wanders the mind wanders; when the breath is steady the mind is steady" comes in the second chapter of that text.

Tirumular, too, sings of breath. The sense of one of his verses is that the body does not weaken for one who knows how to regulate the rising and falling breath; he believed that the practice of attending to the breath guards one's span of life. (Here I give only the sense, without quoting the lines directly.)

✦ Did you know?

The Eastern Christian tradition has an old practice called the "Jesus Prayer": "Lord Jesus Christ, Son of God, have mercy on me, a sinner." Monks such as those of Mount Athos joined it to the breath, saying the first half on the in-breath and the second half on the out-breath. Much writing on it is gathered in the collection called the *Philokalia*.

6.4 Six Breaths a Minute

In 2001 a team led by the Italian physician Luciano Bernardi published a remarkable study in the Christmas issue of the *British Medical Journal*. When participants recited the rosary (the Ave Maria) in Latin, and when they chanted the mantra "Om mani padme hum," their breathing slowed naturally to about six breaths a minute. At that pace the natural rhythms of heart rate, blood pressure and breathing fell into step with one another, and the sensitivity of the *baroreflex*, which controls blood pressure, increased.

The study is small, and it must not be exaggerated. But it points to a beautiful truth: slow, rhythmic breathing has been naturally present in the prayer traditions of many religions. A slow, long out-breath slows the heart through the vagus nerve; this is a well-known effect in studies of *heart rate variability*. A voluntary outer movement (controlling the breath) touches an involuntary inner movement (the heart). The "bridge" of the first chapter appears here before our eyes.

A caution: intense practices such as long breath-retention (*kumbhaka*) and rapid bhastrika can be dangerous in heart disease, high blood pressure, pregnancy and similar conditions; they should be done only under a qualified teacher. Breath-holding exercises under water can be fatal.

6.5 Weeping and Sighing

Breath is not only physiology; it is also the language of emotion. In fear the breath quickens; in grief it becomes a sob; in relief it goes out as a long sigh. A newborn's first cry is its first deep breath; it opens the lungs. On the deathbed, the last breath. Human life begins with a breath and ends with a breath.

The Bible knows this emotional side of breath too. At Lazarus' tomb Jesus "groaned in the spirit, and was troubled" (John 11:33) and wept. Healing a man who was deaf and had a speech impediment, "looking up to heaven, he sighed, and saith unto him, Ephphatha" (Mark 7:34). Paul writes that when we do not know what we should pray for, "the Spirit itself maketh intercession for us with groanings which cannot be uttered" (Romans 8:26). Where words run out, breath itself becomes prayer.

In the yogic *bhramari* pranayama a sound like the humming of a bee is made on the out-breath; practitioners say it calms the mind. This agrees with the scientific fact that a long out-breath brings calm. One who learns the language of breath learns to speak in a new way with his own emotions, and with God.

6.6 The Breath of the Spirit

The Bible's story of breath begins in Eden and goes far. Ezekiel saw a valley full of dry bones. God said: "Behold, I will cause breath to enter into you, and ye shall live" (Ezekiel 37:5). The bones came together and flesh covered them, but only when the breath came did they stand up.

When the risen Jesus appeared to his disciples, "he breathed on them, and saith unto them, Receive ye the Holy Ghost" (John 20:22). The very act by which God breathed into the first man in Eden! It was the beginning of a new creation. On the day of Pentecost the Spirit came down "as of a rushing mighty wind" (Acts 2:2).

Pranayama is the practice of a man handling his own breath. What the Bible speaks of is the grace of God breathing his own breath into man. The two need not conflict: to breathe slowly and grow calm is a good bodily habit. But with that breath the Christian believer says: “Let every thing that hath breath praise the LORD” (Psalm 150:6). Breath is not only the object of a practice; it is an instrument of praise.

6.7 Polluted Air and the Justice of Breath

When we speak of breath we must also speak of the air we breathe. According to estimates of the World Health Organization, outdoor air pollution and indoor cooking smoke together cause millions of premature deaths in the world each year. Village women cooking in the smoke of wood stoves, and poor families living near factories, carry most of this burden. Breath is free for all; but clean breath is not equally available to all.

The duty to protect the breath God breathed into man does not stop at individual practice; it is also social justice. “The earth is the LORD'S, and the fulness thereof” (Psalm 24:1). The air included.

At the individual level there are some simple ways too: stopping smoking (it harms not only one's own lungs but the lungs of nearby children), ventilating the kitchen at home, planting trees, not burning rubbish needlessly. These are small acts; but each is an act of love that protects another's breath. The one who learns pranayama, and the one who sings “Let every thing that hath breath praise the LORD,” must also care about his neighbour's breath.

For Reflection

1. How many times today did you notice your breath? What is it like in moments of anxiety?
2. "Let every thing that hath breath praise the LORD" (Ps 150:6). Can you suggest one simple habit for turning breath into praise?
3. The scene of breath coming into dry bones (Ezek 37): to which "dried-up" part of your life does it speak?

Comparison

	Science	The Bible	Yogic tradition
Engine of breath	Diaphragm; CO_2 sensors	"The breath of life" (Gen 2:7)	Prana
Slow breathing	Vagal stimulation, HRV, baroreflex	Rosary, Jesus Prayer	Pranayama (YS 2:49)
Exercise	Higher VO_2max	—	Pranayama after asana
Spiritual meaning	—	The breath of the Spirit (John 20:22)	The way to a steady mind
Caution	Dangers of intense breath-holding	—	A teacher's guidance is essential

✦ ✦ ✦

CHAPTER 7

Food and Fasting



Eating and abstaining: two lessons for body and soul.

7.1 Eating in Measure

In his chapter on medicine Valluvar wrote a simple but profound line:

> "Marundhena vendaavaam yaakkaikku arundhiyadhu atradhu potri unin." (Kural 942)

The one who eats only after knowing that what he ate has been digested needs no medicine at all. The next couplet says: "Atraal alavarindhu unga aqdhudambu petraan nedidhuykkum aaru" (943): to eat in measure after

digestion is the way to preserve the body long. This counsel of two thousand years ago agrees with the basic message of today's nutrition science.

The Bhagavad Gita speaks of the same balance: yoga is not for one who eats too much nor for one who does not eat at all, not for one who sleeps too much nor for one who is always awake (6:16); for one who is moderate in food and recreation, in actions, in sleep and in waking, yoga destroys sorrow (6:17). The word *yukta*, fitting measure, is a key to the whole idea of this book.

7.2 The Journey of Food Into the Body

When we put a mouthful of rice into our mouth, the outer movement ends and the inner movement begins. Chewing is voluntary; from the moment of swallowing almost everything is involuntary: the waves of the gullet, the acid of the stomach, the enzymes of the pancreas, the bile of the liver, the absorption of the small intestine, the microbes of the large intestine. We command none of it.

Yet our outer choices deeply affect this inner journey. Food rich in fibre becomes food for the gut microbes; from it they make *short-chain fatty acids*, which studies show help the health of the gut wall. Too much sugar and highly processed food are linked over the long term to the risk of metabolic disease. Our grandmothers' diet of pearl millet, finger millet, greens and lentils, in the right measure, is very close to many of today's recommendations.

Daniel and his friends refused the palace food of the king of Babylon and asked: “Prove thy servants, I beseech thee, ten days; and let them give us pulse to eat, and water to drink” (Daniel 1:12). After ten days their faces looked fairer than the others'. This is not a nutrition study; their first purpose was faithfulness. But it shows beautifully the dignity of simple food.

7.3 Fasting: Rest for the Inner Machine

Even when we are not eating, the inner machine does not stop; it switches to other work. After some hours without food the liver releases glucose from its store of glycogen; as time goes on the body begins to use more fat, and ketones are formed. Physiology tells us that in the intervals without food a cleansing wave called the *migrating motor complex* sweeps through the small intestine.

In 2016 the Japanese scientist Yoshinori Ohsumi was awarded the Nobel Prize in Medicine for explaining, in yeast cells, *autophagy*, the process by which cells break down and recycle their damaged parts. Lack of food is one of the factors that triggers this process. But caution is needed here: claims on social media that "if you fast, the body cleans itself and heals every disease" have not been proven in humans. Studies of intermittent fasting mostly show that for weight loss it gives roughly the same benefit as ordinary calorie restriction. People taking diabetes medicine, pregnant women, children and those with eating disorders should not fast without medical advice.

✦ Did you know?

The English word "breakfast" means "breaking the fast" (break + fast). Every night's sleep is a small fast; every morning meal is a small feast that breaks it. The Tamil phrase *nonbu thirattal*, "opening the fast," says the same.

7.4 Fasting in the Bible

In the Bible fasting is not a health exercise; it is the turning of the whole person toward God. We read that Moses, Elijah and Jesus each fasted forty days. When Jesus was hungry in the wilderness the tempter told him to turn stones into bread; he answered: "Man shall not live by bread alone, but by every word that proceedeth out of the mouth of God" (Matthew 4:4).

The prophet Isaiah reveals the true meaning of fasting: the fast God desires is to loose the bands of wickedness, to undo the heavy burdens, to share your bread with the hungry, and to bring the poor that are cast out into your house (Isaiah 58:6–7). Going without food must become a school for remembering the brother who lives with hunger. If the food saved by fasting does not reach the poor, in Isaiah's eyes it is not a complete fast.

Jesus warned: do not put on a sad face so as to be seen by men; fast so as to be seen by your Father who is in secret (Matthew 6:16–18). Fasting is not a spectacle; it is an inner conversation.

7.5 The School of Hunger

Hunger is not only the language of the stomach. When the stomach is empty a hormone called *ghrelin* is released and sends a hunger signal to the brain; *leptin*, secreted by fat tissue, informs the brain about long-term energy stores. It is commonly said that the feeling of fullness arrives fully only about twenty minutes after eating; that is why those who eat slowly mostly eat less. (This time is approximate and varies from person to person.)

But hunger has a spiritual side too. When the people of Israel walked forty years in the wilderness, God gave them manna. Deuteronomy gives its purpose: "And he humbled thee, and suffered thee to hunger ... that he might make thee know that man doth not live by bread only, but by every word that proceedeth out of the mouth of the LORD doth man live" (Deuteronomy 8:3). This is the very verse Jesus quoted in the wilderness. Hunger teaches a person his dependence.

On the other hand, hundreds of millions in the world live with a hunger they did not choose. The one who chooses to fast and the one who starves with no other option are not in the same condition. When Jesus fed the five thousand, the first thing he said was, "Give ye them to eat" (Mark 6:37). One who learns in the school of hunger learns to feed the hungry.

7.6 The Fasting Traditions of Tamil Soil

In Tamil Hindu families fasts are part of life: Ekadasi, Sashti, Pradosham, the mandala fast before Sabarimala, the Saturdays of Purattasi. Many fasts add to the reduction of food other disciplines such as celibacy, abstaining from alcohol and tobacco, gentle speech and bathing before dawn. The yogic word *tapas* means "heat" or "intense effort": a discipline that burns the body to purify the mind. Patanjali places tapas among his *niyamas* (Yoga Sutra 2:32).

Tamil Christians too have Lent, the Good Friday fast, and in some churches a weekly prayer-and-fasting day. This shared practice opens a conversation: can one reach God by subduing the body? The answer of the Christian faith: fasting is not a way to earn God's love; it is a way to express our hunger, our longing, before a love already given. "Whether therefore ye eat, or drink, or whatsoever ye do, do all to the glory of God" (1 Corinthians 10:31). Eating is worship; not eating is worship too.

7.7 The Community Around the Table

Food nourishes not only the body but relationships. In the Gospels Jesus is often at table: he eats with tax collectors and sinners, invites himself to Zacchaeus' house, feeds five thousand, breaks bread at the Last Supper. His enemies accused him of being "a man gluttonous" (Matthew 11:19). Whom we eat with is a declaration of whom we accept.

In Tamil society the communal meal, the *panthi*, has for centuries also been a place of caste division: separate rows, separate tumblers. In 1867 Vallalar Ramalinga Adigal began the Sathya Dharma Salai at Vadalur and fed everyone without regard to caste; it is said that the hearth he lit there still burns unquenched today. His line, "Whenever I saw the withered crop, I withered," is the Tamil summit of compassion for the hungry. A society where all sit around one table is a healthy social body.

For Reflection

1. How can you practise this week the counsel of Kural 942, to eat only after what you ate has been digested?
2. According to Isaiah 58, for whom should your fasting become food?
3. Is fasting "a way to earn love" or "a way to express longing"? What has your experience been?

Comparison

	Science	The Bible	Yogic tradition
Measure	Balanced diet, fibre	All to God's glory (1 Cor 10:31)	Yukta ahara (Gita 6:17)
Physiology of fasting	Glycogen, ketones, autophagy	—	Tapas (YS 2:32)
Purpose of fasting	Metabolic studies (with limits)	Turning to God, justice (Isa 58)	Purity of mind, discipline
Model	—	Daniel's simple food; Jesus' forty days	Vratas (vows)
Caution	Medical advice for the sick	Not for show (Matt 6:16)	Immoderation hinders yoga (Gita 6:16)

✦ ✦ ✦

CHAPTER 8

One Body, Many Members

✦



The eye cannot say to the hand, "I have no need of thee."

8.1 Thirty Trillion Cells

How many cells are there in a human body? According to an estimate made in 2013 by the team of the Italian researcher Eva Bianconi, about 37 trillion. In 2016 the Israeli researcher Ron Milo and his colleagues calculated about 30 trillion for a "reference man," the majority of them red blood cells. These are approximate numbers, but the vastness of the scale is clear.

Each of these cells carries the same genome; yet one serves as a nerve cell, one as a muscle cell, one as a bone cell. Each does its own work, yet none lives for

itself alone. When a cell breaks its controls and begins to multiply for itself, heedless of the body's good, its name is cancer. The health of the body is a community in which trillions of members cooperate for the common good.

8.2 Paul's Parable of the Body

There were divisions in the church at Corinth. Some boasted of their spiritual gifts; some thought themselves useless. Paul showed them the human body:

> "For as the body is one, and hath many members, and all the members of that one body, being many, are one body: so also is Christ." (1 Corinthians 12:12)

Then he draws a humorous scene: if the foot says, "Because I am not the hand, I am not of the body," is it therefore not of the body? If the whole body were an eye, where were the hearing? (12:15–17). "And the eye cannot say unto the hand, I have no need of thee: nor again the head to the feet, I have no need of you" (12:21).

8.3 The Members That Seem Weaker

The deepest line of Paul's parable is this: "those members of the body, which seem to be more feeble, are necessary" (1 Corinthians 12:22). In the language of this book: without the kidneys, liver and pancreas, labouring unseen, unpraised and unceasing within, a beautiful face and strong hands could not live a single day.

Then he says: "And whether one member suffer, all the members suffer with it; or one member be honoured, all the members rejoice with it" (12:26). When a thorn pierces the foot, the eye notices, the hand removes it, the whole body stops to help.

Dr Paul Brand explained this through pain. When leprosy damages the nerves and destroys the sense of pain, the patient does not feel his wound, so his hands and feet are injured again and again and destroyed. Pain is the voice of one member crying to the others, "Help me." Brand called it "the gift nobody wants."

✦ Did you know?

The cells of our body are renewed at different speeds. The cells of the gut lining are replaced in a few days, red blood cells in about 120 days; but most of the brain's neurons last a lifetime. The popular claim that "the whole body becomes new every seven years" is not true.

8.4 Caste and the Body

On Indian soil this parable has a painful history. The Purusha Sukta of the Rig Veda (10.90) sings of the four *varnas* arising from the body of a cosmic man: the Brahmin from his mouth, the Rajanya (Kshatriya) from his arms, the Vaishya from his thighs, the Shudra from his feet. Later this bodily image was used to justify a hierarchy of high and low by birth; Dr Ambedkar sharply criticised this ranking. It must also be remembered here that within the Hindu tradition many sages and reformers opposed inequality based on birth.

Paul's parable runs in the opposite direction. In his body, greater honour is given precisely to the members that "seem weaker" (12:23–24). In the same chapter he says: "For by one Spirit are we all baptized into one body, whether we be Jews or Gentiles, whether we be bond or free" (12:13). The foot is not lower; without the foot the body cannot walk. The feet that labour in the salt pans and the mouth that teaches in the classroom are equal members of one body.

The sad truth: caste has entered even Indian churches: separate cemeteries, separate seating, separate marriage circles. Paul's parable calls us first to judge ourselves.

8.5 Immunity: the Body's Guard

The finest example of the body's unity is the *immune system*. As soon as a small wound occurs, the blood vessels at that spot widen; white cells rush there; some swallow germs; some make *antibodies*; some become "memory cells" that quickly recognise the same germ if it comes again. Vaccines make use of this power of memory.

The basic skill of this system is to tell "self" from "non-self." When it fails there are two kinds of danger: being unable to resist a foreign germ, or attacking one's own organs as an enemy. The second is called *autoimmune disease*: some kinds of arthritis and type 1 diabetes are among them.

This is a deep parable for society. When a society begins to see its own members as strangers and to attack them, it suffers from autoimmune disease. Caste fanaticism, religious hatred, linguistic chauvinism: these are the autoimmune diseases of the social body. Paul warned the Galatians: "But if ye bite and devour one another, take heed that ye be not consumed one of another" (Galatians 5:15). A healthy body recognises and protects its members; so does a healthy society.

8.6 Christ the Head

Paul takes the parable one step further. In the letter to the Ephesians Christ is the "head" of that body: "From whom the whole body fitly joined together and compacted by that which every joint supplieth, according to the effectual working in the measure of every part, maketh increase of the body unto the edifying of itself in love" (Ephesians 4:16).

As in physiology the brain coordinates the body, so Christ coordinates the Church. But this head is not a head that dominates; it is a head that gave itself for its body, a head that wore a crown of thorns. Only when the body is joined to the head are the members joined to one another. To a society split by caste,

language and party, a community living as one body under one head must be a witness.

8.7 The Lesson of a Hospital Ward

Think of the emergency ward of a large government hospital. The surgeon, the anaesthetist, the nurse, the lab assistant, the blood-bank worker, the sanitation worker who cleans the floor: all of them are needed to save one life. If the sanitation worker does not do his work, infection spreads and the surgeon's skill is wasted. But in society who receives honour, and who receives the lowest wage?

During the Covid-19 pandemic the world realised this for a moment: when it looked at who the "essential workers" were, they were mostly low-paid hands: sanitation workers, delivery workers, nurses, farmers. Paul's word appeared as a social truth: "those members ... which seem to be more feeble, are necessary." When the pandemic ended, did we forget again? If we are members of one body, that awareness must not stop at applause; it must become fair wages and respect.

For Reflection

1. Who are the members that "seem weaker" in your church or community? How can greater honour be given to them?
2. "Whether one member suffer, all the members suffer with it" (12:26). Whose pain in your town are you failing to feel?
3. How would you explain respectfully to a friend the difference between the bodily image of the Purusha Sukta and Paul's bodily image?

Comparison

	Science	The Bible	Yogic tradition
Unit	About 30–37 trillion cells (estimate)	Many members (1 Cor 12:12)	Limbs of Purusha (Rig 10.90)
Coordination	Nerve and hormone signals	Christ the head (Eph 4:16)	—
Hierarchy	No hierarchy of worth among organs	More honour to the weaker (12:22–24)	Varna hierarchy in later social reading
Suffering	Pain: a protective signal (Brand)	All suffer together (12:26)	—
Disorder	Cancer: a cell multiplying for itself	Division (1 Cor 1:10)	Caste discrimination

✦ ✦ ✦

CHAPTER 9

The Body Is a Temple

✦



"The heart is the great shrine, the body of flesh the temple" — Tirumular; "your body is the temple of the Holy Ghost" — Paul.

9.1 Tirumular's Turning

Tirumular, the forerunner of the Tamil Siddha tradition, experienced a great change of mind in his life. Two famous verses of the *Tirumantiram* tell of that change:

> "If the body perishes, the life perishes too; nor will one attain firm true knowledge. Knowing the way to nurture the body, I nurtured the body, and so I nurtured the life."

> "Once I held the body to be a defilement; within the body I found the true Reality. Knowing that the Supreme has made his temple within the body, I now cherish and guard the body."

(Tirumantiram, verses 724 and 725 in the traditional numbering)

At first I thought the body *izhukku*, something base; but within the body itself I found the true Reality; knowing that God, the Supreme One, dwells in the body as in a temple, I now cherish and guard it. If the body perishes the life perishes; firm true knowledge cannot be attained; so, knowing the way to nurture the body, I nurtured it, and thus I nurtured the life.

In an age of ascetic traditions that despised the body, this was a bold declaration.

9.2 "The Heart Is the Great Shrine"

Another verse of the Tirumantiram draws the body as a whole temple:

> "The heart is the great shrine, the body of flesh the temple; for the bounteous Lord the mouth is the tower-gate; for those of clear vision the soul is the Sivalingam; the five deceitful senses, the ever-burning jewelled lamps."

The heart is a great temple with its sanctum; the body of flesh is the temple; the mouth is the gate beneath the tower; for the sage of clear understanding the soul itself is the lingam; the five deceiving senses (when subdued) are unquenchable jewelled lamps. These are images every Tamil who goes to a temple understands: the tower, the sanctum, the lamp. Tirumular brings them inside the body.

The Hatha yoga texts of the yogic tradition say the same in another language: the body is the instrument of wisdom; to purify and strengthen it is

the first step of the spiritual journey. The first of Patanjali's niyamas is *shaucha*, purity (Yoga Sutra 2:32).

9.3 Paul's Temple

Corinth was a great port city, notorious for immorality. There some Christians argued, "The body is going to perish; only the soul matters; so whatever one does with the body does not matter." Paul replied sharply:

> "What? know ye not that your body is the temple of the Holy Ghost which is in you, which ye have of God, and ye are not your own? For ye are bought with a price: therefore glorify God in your body, and in your spirit, which are God's." (1 Corinthians 6:19–20)

To the Jews "the temple" meant the temple in Jerusalem, the one place where God's presence dwelt. Paul says: now that presence dwells in the body of every believer. The body is not base; it is holy. But neither is it our property: "ye are not your own."

✦ Did you know?

The Greek word Paul uses for "temple" in 1 Corinthians 6:19 is *naos*. It refers not to the whole temple complex (*hieron*) but to the inner holy place where God's presence dwells. In other words, the body is not merely the temple precinct; it is the sanctuary itself!

9.4 Destroy This Temple, and in Three Days

The image of the temple takes a startling turn in the Gospel. After driving the traders out of the temple in Jerusalem, Jesus said to those who asked for a sign: "Destroy this temple, and in three days I will raise it up." They thought he was

speaking of the building. John explains: “But he spake of the temple of his body” (John 2:19–21).

The body of Jesus is the true temple: in it the fullness of God dwelt bodily (Colossians 2:9). That temple was destroyed on the cross and raised up again on the third day. On that day the veil of the temple in Jerusalem was torn in two from top to bottom (Matthew 27:51). From then on God's presence was no longer shut up inside a building.

That is why Paul can call the bodies of believers temples: the bodies of those joined to the temple that is Christ's body become stones of his temple. “Ye also, as lively stones, are built up a spiritual house” (1 Peter 2:5). Tirumular saw a temple in his own body; the Christian faith sees in one body, broken and risen, the fulfilment of all temples.

9.5 Two Temples: Likeness and Difference

That both traditions call the body a temple is a striking likeness. Both oppose the tendency to despise the body; both call us to cherish and guard it; both see a divine presence within it.

But the differences are subtle too. In Tirumular's view God is already within the body; “within the body I found the true Reality”: discovery, knowing. The line “the soul is the Sivalingam” points to a deep oneness, that the soul itself is a form of God. In Paul's view the Spirit comes in as a gift, “which ye have of God.” The body becomes a temple not by nature but because it was “bought with a price”: the price paid on the cross. In one the body is a temple because God is in its depths; in the other the body is a temple because God bought it at a price and came to dwell in it.

Another difference: Tirumular says the body is an instrument of wisdom, “I nurtured the body, and so I nurtured the life.” Paul says the body is the sacrifice of worship: “present your bodies a living sacrifice, holy, acceptable unto God,

which is your reasonable service" (Romans 12:1). The body is not a ladder to climb; it is an offering to be given.

9.6 Guarding the Temple

If it is a temple, we will keep it clean, repair it when it breaks, and not dump rubbish in it. This image raises practical questions. How do tobacco, alcohol and drugs ruin this temple? The drug habit spreading among young people in Tamil Nadu is like the looting of temples. Overeating and inactivity are a slow decay. Restless toil and sleeplessness are cracks in the temple walls.

But a caution: the idea of "the body as temple" must not turn into the worship of bodily beauty. Today's world of social media turns the body into an object of display and competition. The sick body, the aged body and the differently abled body are equally temples. The dignity of a temple lies not in the colour of its walls but in the presence that dwells within it. Here Tirumular and Paul would speak with one voice: we guard the body in order to honour the One within it.

9.7 A Whip in the Temple

When Jesus entered the temple in Jerusalem, there were people selling cattle and sheep and changing money. He made a whip of cords and drove them out: "make not my Father's house an house of merchandise" (John 2:16). The temple was for prayer; when it became a market for profit, Jesus was angry.

The temple of our body, too, is today in danger of becoming a market. Food companies turn our hunger into profit; advertisements turn our shame about our bodies into profit; phone apps turn our attention and our sleep into profit. Sometimes even yoga and exercise lose their spiritual meaning and become expensive commodities.

The challenge of this chapter: which "traders" have entered the temple of your body? Do you need Jesus' whip to drive them out? As Tirumular said,

only when "the five deceitful senses" are subdued do they become jewelled lamps.

Cleansing the temple is not a one-time event; it is a daily task. Our forebears swept the temple and the house each morning, smeared the floor and drew a kolam. The temple of the body needs such a daily order too: a few minutes of prayer and quiet breathing in the morning, honest labour and measured food by day, a short walk in the evening, and sleep with thanksgiving at night. These are small habits; but as the years pass they rise up as the walls of a temple. As Paul wrote to Timothy, "exercise thyself rather unto godliness" (1 Timothy 4:7): here the Greek word for "exercise" (*gymnazō*) comes from the gymnasium. The spiritual life, too, is a training.

For Reflection

1. Was there a time when, like Tirumular, you thought of the body as "base"? What is your view now?
2. What should the truth "ye are not your own" (1 Cor 6:19) change in your bodily habits?
3. How would you explain to young people the difference between worshipping bodily beauty and honouring the body as a temple?

Comparison

	Science	The Bible	Yogic tradition
Status of the body	A biological system	Temple of the Holy Spirit (1 Cor 6:19)	The body of flesh a temple (Tirumantiram)
Divine presence	—	The Spirit who comes in as a gift	God already dwelling within
Why holy?	—	Because bought with a price	"The soul is the Sivalingam"
Role of the body	—	Living sacrifice, worship (Rom 12:1)	Instrument of wisdom (v. 724)
Guarding it	Avoiding drugs, inactivity	Glorify God (6:20)	Shaucha (YS 2:32)

CHAPTER 10

The Creator's Handiwork

✦



"I am fearfully and wonderfully made."

10.1 The Poem of the Womb

Psalm 139 is one of the most beautiful poems ever written about the human body. Three thousand years ago, before the science of embryology was born, David sang of the hidden work that goes on in the womb:

> "For thou hast possessed my reins: thou hast covered me in my mother's womb. I will praise thee; for I am fearfully and wonderfully made: marvellous are thy works; and that my soul knoweth right well." (Psalm 139:13–14)

The Hebrew word translated "reins" (*kilyah*) literally means "kidneys"; it also signifies a person's inner centre. The "inner machine" of this book is, in this psalm, in the hand of God. The next verses say: "My substance was not hid from thee, when I was made in secret, and curiously wrought in the lowest parts of the earth. Thine eyes did see my substance, yet being unperfect" (139:15–16). The Hebrew word behind "curiously wrought" is related to the craft of weaving: I was knit, I was woven.

10.2 From One Cell to a Human Being

Every human being begins as a single fertilised cell, a *zygote*. It splits into two, then four, then eight. Within a few weeks three layers form, and from them come skin and nerves, muscles, bones and heart, gut and lungs. About three weeks after conception the heart tube begins to beat; the heart is among the first organs of the human body to start working. In other words, the inner machine begins to run before the outer hands are formed. The hands appear as paddle-like buds, and the fingers separate when the cells between them die according to plan (*apoptosis*).

When the embryo's heart starts to beat, the mother may not even know she is pregnant. How true is the line "I was made in secret"! The most important work of our lives began before we knew it, before even our mother knew it.

10.3 The Signature of the Hands

A small wonder in the design of our body: fingerprints. Every person's fingerprints are unique; even identical twins with the same genes have different fingerprints, because tiny differences in the environment of the womb shape them as the skin of the fingers forms. India's Aadhaar identity card records fingerprints and the iris of the eye precisely on the basis of this uniqueness.

A verse of the prophet Isaiah gives this a beautiful reverse side. To Israel, who felt forsaken, God says: "Can a woman forget her sucking child, that she should not have compassion on the son of her womb? yea, they may forget, yet will I not forget thee. Behold, I have graven thee upon the palms of my hands" (Isaiah 49:15–16). Our fingerprints identify us; our name is engraved on the palm of God's hand.

When a Christian reader reads this verse he cannot help thinking of the nail wounds in the palms of the risen Jesus. The hands that formed us are the hands that were wounded for us. The Creator's handiwork is seen not only in the womb but also at Calvary.

✦ Did you know?

The DNA in the nucleus of one human cell, stretched out, would be about two metres long; it is folded and packed inside a nucleus a few micrometres across. The weaving image of "curiously wrought" (Ps 139:15) fits here quite literally.

10.4 Design or Wonder?

When we marvel at the complex structure of the human body an honest question arises: is this God's design, or the result of evolution? This book does not try to settle that scientific debate. Many Christian scientists (such as Francis Collins, who led the Human Genome Project) believe that God creates through natural processes. Others think differently.

But the psalmist's wonder is not diminished by any scientific explanation. To learn how a painting was painted is not to deny the painter; when we know how the colours were mixed, the painter's skill becomes even more astonishing. “The heavens declare the glory of God; and the firmament sheweth his handywork” (Psalm 19:1). Not only the sky: the believer sees every cell as a witness to that handiwork.

10.5 The Wisdom of the Body: Brand's Wonder

In his books Dr Paul Brand often marvelled at the design of the body. As a surgeon, when he tried to transfer the tendons of the human hand to new positions, he wrote that he could never create an arrangement better than the natural one. Bone competes in strength-for-weight with many engineering materials; when it breaks, it knits itself together. Skin is a living, self-repairing covering.

From this wonder Brand drew a spiritual lesson: the Creator has placed a wisdom of self-protection in his creation, and we must cooperate with that wisdom. The labour of the outer machine, rest, food and breath, all that we have seen throughout this book, are forms of that cooperation.

10.6 The Wonder of the Yogic Tradition

The Indian tradition too looked on the body with wonder. Hatha yoga texts say there are 72,000 *nadis* (subtle channels) within the body. This is not an anatomical figure; it is a symbolic, spiritual map that cannot be matched directly with nerves or blood vessels. But it expresses the wonder of seeing the body as a web of countless channels of life.

Tirumular, too, sang of the forming of the child in the womb; in the second *tantra* of the Tirumantiram there are verses on *garbha kriya*, the work of the womb. In his view the shaping of the body in the womb is God's grace. David and Tirumular, though separated by three thousand years and four thousand

kilometres, bow before the womb in the same wonder. The last word of that wonder is thanks: "I will praise thee."

10.7 From Wonder to Responsibility

If the body is God's handiwork, so are the bodies of others. This truth must change the way we see society. Those who look at a differently abled child and say "the result of sin" forget Psalm 139. A society that looks down on dark skin forgets the line "marvellous are thy works." In a country full of advertisements for skin-whitening creams, this is an important message.

God said to Moses: "Who hath made man's mouth? or who maketh the dumb, or deaf, or the seeing, or the blind? have not I the LORD?" (Exodus 4:11). This is a hard verse; not every bodily impairment can be given a theological explanation. But one message of it is clear: no body is outside God's attention; no body is of lesser worth. If our wonder is real, it must become respect and responsibility.

This responsibility begins in the womb. Public health research stresses that the nutrition of the pregnant woman, safe childbirth and the care of a child's first thousand days shape its health for a whole lifetime. Tamil Nadu's government maternity hospitals, nutrition schemes and anganwadis serve the "poem of the womb" in this way. The believer who sings, "Thine eyes did see my substance," must also see with his own eyes the unborn child of the poor mother, and take part in making sure that mother receives food and medical care. Wonder becomes worship; worship becomes service.

For Reflection

1. Stand before a mirror and say Psalm 139:14. What do you feel about your body?
2. Does scientific explanation lessen or increase wonder at God? Why?
3. What does Psalm 139 tell us about the dignity of life in the womb?

Comparison

	Science	The Bible	Yogic tradition
Beginning	One fertilised cell	"Thine eyes did see my substance" (139:16)	Garbha kriya (Tirumantiram)
First movement	Heart tube beating at about 3 weeks	"Thou hast possessed my reins" (139:13)	The entry of prana
Structure	DNA, cell specialisation	"Curiously wrought" (139:15)	72,000 nadis (symbolic)
Response	Wonder, inquiry	"I will praise thee" (139:14)	A sense of grace

CHAPTER 11

The Broken Machine and the Redeemer

✦



Hands that touched broken bodies; a body broken on the cross.

11.1 An Honest Question

So far we have seen the wonders of the body. But walk into any hospital and another truth strikes us in the face. Hearts stop. Kidneys fail. Labouring hands are paralysed by stroke. Children writhe with cancer. However honestly we labour, exercise and eat, in the end every machine stops.

Health books mostly fall silent at this point. They stop at "live well, live long." But true wisdom must face the face of death as well. This chapter is that encounter.

11.2 Ageing: the View of Science

Why does the body age? Science has no single answer; many factors work together. Each time a cell divides, the *telomeres* at the ends of the chromosomes shorten a little; damage accumulates in DNA; the power of the mitochondria, the cells' energy centres, declines; *senescent cells* that have stopped dividing secrete substances that provoke inflammation. A famous review published in 2013 listed nine "hallmarks of aging"; in 2023 the list was expanded further.

Studies show that exercise, good food and sleep help delay many effects of ageing; but nothing stops it. The psalm written by Moses says so honestly: "The days of our years are threescore years and ten; and if by reason of strength they be fourscore years, yet is their strength labour and sorrow; for it is soon cut off, and we fly away" (Psalm 90:10).

11.3 The Bible: the Root of the Breaking

The Bible sees death not as something natural but as a foreign invasion. In Eden God warned: "in the day that thou eatest thereof thou shalt surely die" (Genesis 2:17). After sin: "dust thou art, and unto dust shalt thou return" (3:19). Paul says briefly: "by one man sin entered into the world, and death by sin" (Romans 5:12). "The whole creation groaneth and travaileth in pain together" (Romans 8:22).

But take care: this does not mean that every illness is the punishment of that person's own sin. When the disciples asked about the man born blind, "who did sin, this man, or his parents?", Jesus rejected both (John 9:3). Illness is the common condition of a broken world; to make it a personal guilt is cruelty. The whole book of Job is a long cry against that cruelty.

The yogic and Hindu traditions explain suffering within the framework of karma and rebirth; Patanjali says, "the suffering yet to come is to be avoided" (*heyam duhkham anagatam*, 2:16). The Buddha's first truth, too, is suffering. All human traditions have seen this breaking; the difference lies in the remedy.

✦ Did you know?

Tradition says the beginning of the Christian Medical College, Vellore, is linked to the events of one night: three men asked the young Ida Scudder for help with their wives' childbirth; because custom forbade a male doctor to touch them, all three women died. That experience drove her to study medicine. Later Dr Paul Brand served in the same institution.

11.4 Healing Hands

In the Gospels much of Jesus' ministry took place among broken bodies. He touched the leper, a man no one touched in those days (Mark 1:41). A woman who had suffered from bleeding for twelve years, who "had suffered many things of many physicians, and had spent all that she had," and was no better, touched his garment; he called her "Daughter" (Mark 5:25–34). To the man with the withered hand he said, "Stretch forth thine hand" (Mark 3:5): a hand that could not work became a labouring hand again.

Matthew links these healings to Isaiah's prophecy: "Himself took our infirmities, and bare our sicknesses" (Matthew 8:17). Following this tradition, Christian missionaries built hospitals across India; Ida Scudder, who began the Christian Medical College, Vellore, as a small hospital in 1900, was one of them. Healing stands not at the margin of the gospel but at its centre.

But Jesus did not heal every sick person; and those he healed also died in the end. Lazarus died again. The healings were signs: a foretaste of a greater redemption. That redemption took place on the cross.

11.5 Living With Pain

Not every illness is healed. Among us are many who live with long-term illness, who are in the last stage of cancer, who are in the frailty of old age. To tell them, "If you had enough faith you would be healed," is both unbiblical and cruel. Paul himself had "a thorn in the flesh"; three times he prayed for it to leave. The answer came: "My grace is sufficient for thee: for my strength is made perfect in weakness" (2 Corinthians 12:9).

In India the palliative care movement has grown strong in Kerala; its aim is to relieve pain, protect the patient's dignity and support the family. "To cure sometimes, to care always" is an old saying of medicine. Cicely Saunders, the pioneer of the modern palliative care movement, was a Christian believer; in 1967 she founded St Christopher's Hospice in London.

Jesus did not heal everyone; but he turned away no one who came to him. In Gethsemane he himself asked for his disciples' presence: "My soul is exceeding sorrowful, even unto death: tarry ye here, and watch with me" (Matthew 26:38). Sometimes the greatest medicine we can give to those in pain is our presence.

11.6 The Broken Body

At the Last Supper Jesus took bread, broke it, and said: "This is my body, which is broken for you" (1 Corinthians 11:24). The next day that body was truly broken: the scourging, the crown of thorns, hands and feet pierced by nails. The labouring hands, the carpenter's hands, the healing hands were nailed to the wood.

Physicians have written many articles on the physiology of Jesus' death (an article published in JAMA in 1986 became widely known; some of its conclusions were later debated). There is no consensus among doctors on the exact medical cause of death by crucifixion. But many explain that a crucified man had to struggle even to breathe. He who gave the breath of life that God

breathed into the first man struggled for breath. “And Jesus cried with a loud voice, and gave up the ghost” (Mark 15:37).

Isaiah foretold: “But he was wounded for our transgressions, he was bruised for our iniquities: the chastisement of our peace was upon him; and with his stripes we are healed” (Isaiah 53:5). To redeem a broken humanity, God gave his own body to be broken. The repair of the broken machine did not come from outside; the Repairer himself was broken.

11.7 The Physician and the Redeemer

The Christian faith is not opposed to medicine. Luke the evangelist was a physician; Paul calls him “Luke, the beloved physician” (Colossians 4:14). In the Old Testament it was the prophet Isaiah himself who told them to lay a lump of figs on King Hezekiah's boil (Isaiah 38:21). Prayer and medicine are not enemies; both are channels of God's grace. Jesus himself said, “They that be whole need not a physician, but they that are sick” (Matthew 9:12).

But the physician repairs the damage of the body; the Redeemer redeems the whole person. Medicine postpones death; the Redeemer conquers death. Even the best hands of medicine come at last to a place where they must let go. There, on the edge of the last breath, the One who was broken on the cross says: “I will never leave thee, nor forsake thee” (Hebrews 13:5). This is the doorway to the last chapter of this book.

For Reflection

1. Is there a habit around you of seeing illness as “the punishment of personal sin”? What answer does John 9:3 give?
2. Jesus “touched” the leper. Whom does our society hesitate to touch today?
3. “The Repairer himself was broken”: how does this truth change the way you see your own brokenness?

Comparison

	Science	The Bible	Yogic tradition
Breaking	Hallmarks of ageing, disease	Sin and death (Rom 5:12)	Karma, duhkha (YS 2:16)
Personal guilt?	No: genes, environment, age	No (John 9:3)	Explained by karma doctrine
Healing	Medicine (with limits)	The hands of Jesus (Mark 1:41; 5:34)	Yoga therapy (with limits)
Final remedy	—	The broken body (Isa 53:5; 1 Cor 11:24)	Moksha, release from the cycle of births

✦ ✦ ✦

CHAPTER 12

The Risen Body



An empty tomb, a new morning, an imperishable body.

12.1 The Morning of the Third Day

Very early on the first day of the week, some women went to the tomb to anoint Jesus' body with spices. They went looking for a body; what they found was an empty tomb. “Why seek ye the living among the dead? He is not here, but is risen” (Luke 24:5–6).

This is the centre of the Christian faith. Paul says it plainly: “if Christ be not raised, your faith is vain” (1 Corinthians 15:17). The resurrection is not merely a spiritual parable; it is an event that concerns the body. To the disciples who

thought they "had seen a spirit," the risen Jesus said: "Behold my hands and my feet, that it is I myself: handle me, and see; for a spirit hath not flesh and bones, as ye see me have" (Luke 24:39). Then he ate broiled fish before them (24:42–43). He told Thomas to touch the wounds of his hands (John 20:27). Hands that laboured, hands that were nailed, now risen hands.

12.2 Sowing Incorruption

The Corinthians asked: "How are the dead raised up? and with what body do they come?" Paul answered with a farmer's parable. The seed sown in the ground comes to life only if it dies; what is sown is bare grain, what rises is a whole plant. So also:

> "It is sown in corruption; it is raised in incorruption: it is sown in dishonour; it is raised in glory: it is sown in weakness; it is raised in power: it is sown a natural body; it is raised a spiritual body." (1 Corinthians 15:42–44)

A "spiritual body" does not mean a bodiless spirit; it is an imperishable body, wholly given life by the Spirit. Seed and plant are one, yet the change is immense. The Christian hope is not the soul's escape from the body; it is the redemption of the body itself. "Who shall change our vile body, that it may be fashioned like unto his glorious body" (Philippians 3:21).

12.3 Hands That Labour to the End

The hope of the resurrection does not tell us to neglect life in this world. Paul ends his long chapter on the resurrection like this: "Therefore, my beloved brethren, be ye stedfast, unmoveable, always abounding in the work of the Lord, forasmuch as ye know that your labour is not in vain in the Lord" (1 Corinthians 15:58).

Notice: the conclusion of the resurrection is "work"! If the body will rise, then labour done with the body has meaning. The hand that fed the sick, the hand that taught a poor child, the hand that toiled in the salt pans, the carpenter's hand that drove nails into wood: nothing done in the Lord is wasted. As we said in the first chapter, labour is part of creation; now it is also a seed of the new creation.

On the shore of the Sea of Galilee the risen Jesus kindled a fire for his disciples, prepared fish and bread, and called them: "Come and dine" (John 21:12). For fishermen who had cast their nets all night and caught nothing, the risen Lord cooked breakfast. Could there be a resurrection scene closer to the fishermen of Thoothukudi than this?

✦ Did you know?

The early Christians buried their dead rather than burning them, and called the burial ground *koimētērion*, "the sleeping place." From it comes the English word "cemetery." The hope that death is the sleep of a long night and the resurrection its morning lives in the word itself.

12.4 Within, Even Now: the Indwelling Spirit

The resurrection is not only in the future; its power is already at work within the believer. Paul writes: "But if the Spirit of him that raised up Jesus from the dead dwell in you, he that raised up Christ from the dead shall also quicken your mortal bodies by his Spirit that dwelleth in you" (Romans 8:11).

Let us return to the beginning of this book. In our body an inner machine labours unceasingly without our permission; an outer machine labours at our will. The Christian faith speaks of a third, still deeper "inner movement": the Holy Spirit dwelling within. As the heart keeps our body alive, the Spirit renews "the inward man ... day by day" (2 Corinthians 4:16). The outward man

perishes: age, illness, weariness. But the inward man is renewed. The body's heart will one day stop; the Spirit's inner work will never stop.

That inner work bears fruit outside: "But the fruit of the Spirit is love, joy, peace, longsuffering, gentleness, goodness, faith, meekness, temperance" (Galatians 5:22–23). As outer labour nourishes the inner machine, here the Spirit within gives the outer hands the labour of love. The twofold movement of this book turns round here and is made complete.

12.5 Moksha and Resurrection: a Respectful Comparison

In the yogic and Vedantic traditions final liberation (*moksha*, *kaivalya*) is mostly explained as release from the body and from the cycle of births. The final goal of Patanjali's Yoga Sutras is *kaivalya*: the isolation of *purusha* (pure consciousness) from *prakriti* (nature, including the body). In the Saiva Siddhanta background of Tirumular, the final state is inseparable union with Siva. These visions are the fruit of a deep search, and we honour them.

The Christian hope goes in another direction: not liberation from the body, but the liberation of the body. "We ourselves groan within ourselves, waiting for the adoption, to wit, the redemption of our body" (Romans 8:23). The final vision is a new heaven and a new earth, where "God himself shall be with them, and be their God. And God shall wipe away all tears from their eyes; and there shall be no more death, neither sorrow, nor crying, neither shall there be any more pain" (Revelation 21:3–4). To wipe away tears there must be a face; and that face is the face of a body. Creation is not rejected; it is renewed.

12.6 I Am the Life

Throughout this book we have spoken of life: the beating heart, the breathing lungs, the labouring hands, the resting body, the eating mouth, the unity of many members. Science explains the movements of life with astonishing precision; but it has no answer to the question, "What is life for?" The yogic

tradition offers deep practices for ordering the life-force, and many of them may benefit body and mind. But for death, the final problem of the broken machine, no practice gives a final answer.

Jesus Christ did not merely show a way to life; he declared himself to be life: "I am the way, the truth, and the life" (John 14:6). "I am come that they might have life, and that they might have it more abundantly" (John 10:10). Beside the tomb of Lazarus: "I am the resurrection, and the life: he that believeth in me, though he were dead, yet shall he live" (John 11:25).

The tireless machine within will one day rest. The labouring hands without will one day be folded and laid upon the breast. But the breath God breathed is not wasted; the Christian hope is that those joined to the body that was broken and rose again will rise with a new body. To the mother labouring in the salt pans, to the fisherman casting his net at sea, to the old man struggling for breath on a hospital bed, the final message of this book is one: your body is the Creator's handiwork; your labour is worship offered to him; he bore your brokenness; he himself is the fullness of your life.

12.7 Conclusion: Three Lenses, One Face

We began this book with three lenses. The lens of science showed the movements of the body with astonishing clarity: nerves, hormones, cells, cycles. We are grateful for it; its discoveries save millions of lives. The lens of the yogic tradition showed an ancient wisdom for ordering breath, body and mind; many of its practices still help many people today, and we honour the search of that tradition. The lens of the Bible showed the body's beginning, its breaking and its final redemption.

But according to the witness of the Christian faith, when we look through the third lens we do not see only an idea; we see a face. "For now we see through a glass, darkly; but then face to face" (1 Corinthians 13:12). That face is the face of Jesus Christ, who laboured as a carpenter, grew weary and slept, hungered

and fasted, struggled for breath and died, and rose on the third day. The tireless machine within and the labouring hands without find their meaning in him. He is the true life that our body and soul need.

For Reflection

1. Is it new to you that resurrection is not "the soul escaping the body" but "the body itself being redeemed"? How does it change the way you see your body?
2. Which fruit of the Spirit (Gal 5:22–23) most needs to show itself now in your "outer hands"?
3. How would you share Jesus' declaration "I am the life" (John 14:6) with a dear Hindu friend, with respect and love?

Comparison

	Science	The Bible	Yogic tradition
After death	Beyond the limits of science	Resurrection of the body (1 Cor 15:42–44)	Kaivalya / moksha
Final state of the body	—	A glorious body (Phil 3:21)	Release from prakriti
Inner movement now	Autonomic nervous system	The indwelling Spirit (Rom 8:11)	Prana, sadhana
Source of life	Biochemical processes	"I am the life" (John 14:6; 10:10)	Brahman / Sivam
Final vision	—	New heaven, new earth (Rev 21:4)	End of the cycle of births

Sources

The primary texts used in this book, and further works for those who wish to read more:

The Bible

- The Holy Bible, Authorized (King James) Version; English Standard Version (Crossway, 2001). Tamil edition quotations: Tamil Union Version (Bible Society of India).

Tamil and Indian Source Texts

- Tirumular, *Tirumantiram* (Tenth Tirumurai).
- Tiruvalluvar, *Tirukkural* (chapters on Medicine and on Farming).
- *Bhagavad Gita* (chapters 2, 3, 6).
- Patanjali, *Yoga Sutras*.
- Svatmarama, *Hatha Yoga Pradipika*.
- *Taittiriya Upanishad*; *Rig Veda* 10.90 (Purusha Sukta).
- Swami Vivekananda, *Karma Yoga* (1896).
- Swami Satyananda Saraswati, *Yoga Nidra* (Bihar School of Yoga).

Christian and Comparative Works

- Paul Brand & Philip Yancey, *Fearfully and Wonderfully Made* (Zondervan, 1980).
- Paul Brand & Philip Yancey, *In His Image* (Zondervan, 1984).

- Paul Brand & Philip Yancey, *Pain: The Gift Nobody Wants* (HarperCollins, 1993).
- J.-M. Déchanet, *Christian Yoga* (Harper & Row, 1960; French original *La voie du silence*, 1956).
- William J. Broad, *The Science of Yoga: The Risks and the Rewards* (Simon & Schuster, 2012).
- Francis S. Collins, *The Language of God* (Free Press, 2006).
- B. K. S. Iyengar, *Light on Yoga* (1966).
- *The Philokalia*, tr. G. E. H. Palmer, P. Sherrard & K. Ware (Faber & Faber, 1979–).

Scientific Sources

- Walter B. Cannon, *The Wisdom of the Body* (W. W. Norton, 1932).
- J. N. Morris et al., "Coronary heart-disease and physical activity of work," *The Lancet* 262 (1953): 1053–57, 1111–20.
- L. Bernardi et al., "Effect of rosary prayer and yoga mantras on autonomic cardiovascular rhythms," *BMJ* 323 (2001): 1446–49.
- B. K. Pedersen & M. A. Febbraio, "Muscle as an endocrine organ," *Physiological Reviews* 88 (2008): 1379–1406.
- World Health Organization, *Guidelines on Physical Activity and Sedentary Behaviour* (2020).
- L. Xie et al., "Sleep drives metabolite clearance from the adult brain," *Science* 342 (2013): 373–77.
- E. Bianconi et al., "An estimation of the number of cells in the human body," *Annals of Human Biology* 40 (2013): 463–71.
- R. Sender, S. Fuchs & R. Milo, "Revised estimates for the number of human and bacteria cells in the body," *PLOS Biology* 14 (2016): e1002533.

- C. López-Otín et al., “The hallmarks of aging,” *Cell* 153 (2013): 1194–1217; updated *Cell* 186 (2023).
- Matthew Walker, *Why We Sleep* (Scribner, 2017).
- Arthur C. Guyton & John E. Hall, *Textbook of Medical Physiology* (various editions).

Acknowledgements

My first thanks go to the readers who read my two earlier books with love and asked, "What next?" Their questions were the fuel for this new journey.

The fishermen, salt-pan workers, construction workers and farmers of Thoothukudi: your labouring hands were the first textbook of this book. This book is a small salute to your sweat.

My thanks to the doctor friends who explained physiology and medicine simply, to the yoga teachers who explained the subtleties of the yogic tradition with respect, to my comrades of the Jai Bhim Educational Charitable Trust, and to my family who always stand by me.

Above all, all glory to Jesus Christ, who made me fearfully and wonderfully, who bore my brokenness, and who gave me life.

With gratitude,

Neelam P.J. Pradeep Jesudas, B.Th

Author

A Tireless Machine Within... Labouring Hands Without!

Even while you sleep, your heart beats, your lungs breathe, your kidneys filter. When you wake, your hands and feet begin to labour. Two movements in one body: one that never rests, and one that obeys your will. Modern science confirms that when the hands outside labour honestly, the machine inside grows strong.

From the conductors of London's buses to the leprosy hospital of Vellore, from the karma yoga of the Bhagavad Gita to Tirumular's “the body is a temple,” from the first breath in Eden to the empty tomb: this book explores the secret of the human body through three lenses, science, the Bible and yoga.

At the end of a journey through labour, rest, breath, food, fasting and the unity of the community, this book respectfully presents the answer that the broken and dying body finds in Jesus Christ.

“I will praise thee; for I am fearfully and wonderfully made.”

— Psalm 139:14

Neelam P.J. Pradeep Jesudas, B.Th

JAI BHIM EDUCATIONAL CHARITABLE TRUST